

Active Proxy Dashboard: Binding Physical Referents and Abstract Data Representations in Situated Visualization through Tangible Interaction

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Figure 1: Left) The conceptual model of the Active Proxy approach consists of the user, active proxy, and abstract data representation such as a dashboard. The physical affordance of the model attracts the user (i), which then actively manipulates the model. The physical manipulation is mapped to data visualization operations (ii) that are observed by the user (iii) and is understood as binding between referent and data (iv). The potential applications of active proxy dashboards include analysis in a shared workspace (middle) and individual analysis in a personal workspace (right).

ABSTRACT

Myriad systems have been proposed in recent years involving situated visualization with tangible scale models as proxies for physical referents. Most of these focus on static placements of scale models that mimic their real-world arrangement. While such static placement is useful to show the spatial context of referents, it does not take full advantage of the potential interaction affordances of the proxies. Our approach, Active Proxy, binds the data representation and referent through physical manipulations of scale models. We introduce a conceptual design space defining four quadrants as combinations of two dimensions: spatial to abstract representation; and passive to active proxy. Our focus is the novel quadrant defined by active proxies and abstract representations. Designing active proxy techniques is non-trivial as users are accustomed to common

interaction modalities. This paper presents an initial exploration toward a better understanding of the active proxy concept, and designs of active proxy systems.

CCS CONCEPTS

• **Human-centered computing** → **Visualization techniques; HCI theory, concepts and models.**

KEYWORDS

tangible interaction, situated visualization, proxsituated visualization, active proxy, data exploration

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1 INTRODUCTION

Our research builds on situated visualizations that combine tangible scale models of physical objects with digital data representations [4, 17]. Situated visualization and analytics is a data visualization approach that incorporates the physical objects or spaces to which the data refers in the analysis process [2, 3, 24, 26, 27]. These objects or spaces are known as *physical referents* [27]. In situations where the referents are inaccessible they can be represented using proxies, in scenarios known as *prox-situated visualization* [16]. For example, buildings (referent) can be represented using scale models (referent proxy). Existing work involving situated visualization with referent proxies tends to arrange the scale models in fixed positions according to their actual arrangement in the real world. For instance, building scale models are placed on pre-determined positions on a tabletop map.

The fixed model arrangement is a reasonable design decision as the layout of these models reflects relative distances between the represented objects. The limitation of such techniques from the tangible interaction perspective is that the tactility of the models is barely utilized in the exploration process. Tangible interaction has been demonstrated to be beneficial in facilitating learning [20]. However, the tangible interaction of physical scale models in situated visualization is still in its infancy with only a few existing works advancing this concept. This research gap motivates our project. We ask two main questions. 1) Do the benefits of tangible interaction translate to interactive situated visualizations that rely on active manipulation of referent proxies such as scale models? 2) How can an active proxy system be designed to be intuitive, taking into account that users are accustomed to traditional input devices and techniques?

In this late-breaking work paper, we present our initial exploration of a tangible-driven approach for situated visualization using proxies of physical referents. We refer to this technique as Active Proxy. We first propose conceptual quadrants that can position existing and future work, as well as our own. These quadrants are the product of two axes: 1) passive vs. active proxy, and 2) spatial vs. abstract representation. Our work focuses on active proxies for understanding data using abstract representations such as a charts dashboard. Our motivating argument is that performing physical manipulations of scale models to explore data forms a strong conceptual binding between the referent and the data in the user's mind.

The term “binding” is inspired by theories of learning [15], where binding is considered as a process of associating features of different entities in one's memory. In situated visualization, physical referent and data representations can be considered the main components and the association between the two is important to the analysis. The embedded data representation [27] concept binds the charts and physical referent by their physical and perceptual locations, creating an association through minimized spatial distance. Our idea of active proxy fosters this association but the main focus is on tangible interactions, rather than spatial distance. By actively manipulating scale models of the physical referents to explore complex abstract data representation (e.g., to perform filtering or selection), we expect to achieve a stronger binding than using a

passive proxy combined with external input devices. We elaborate further on this in section 2.

This paper paves the path to several potential contributions. At a conceptual level, we propose the idea of an active proxy that is expected to create a binding between the physical referent proxy and the data by enforcing tangible interaction. We propose conceptual quadrants describing the type of proxy and the type of data representation. From the application standpoint, we present an initial exploration of an Active Proxy Dashboard prototype system for *prox-situated visualization* [16] of energy usage data of campus buildings. Section 3 describes this prototype. We conclude this paper in section 4 with a brief reflection and description of future work.

2 RELATED WORK AND CONCEPTUAL QUADRANTS

The active proxy concept is highly related to the notion of tangible interfaces replacing digital GUI elements with physical tangible UI, suggested as early as 1993 [22], and derivatives such as graspable interfaces [5] and tangible bits [8]. Our active proxy concept, however, does aim to make abstract UIs tangible, but rather make tangible proxy representations of real-world objects. These proxies not only provide physical and spatial information about real-world objects but also act as interfaces to perform data exploration. There is existing work that demonstrates data exploration using the manipulation of physical tokens with abstract geometry (e.g., a cylinder) [9, 10, 23]. In their work, data operations (e.g., filtering, encoding) are mapped to the physical location and arrangement of physical tokens. Using only basic geometry as tokens means that physical information important for situated visualization is not provided. Our research aims to connect the fields of situated visualization and tangible interaction by considering the manipulation of physical referent proxies for data exploration. Thus, our envisioned active proxy approach maintains the geometrical property of the referent by using scale models. Our investigation revealed the need for a conceptual design space to describe existing and future work. We propose conceptual quadrants consisting of 1) passive proxy vs. active proxy and 2) spatial representation vs. abstract representation factors.

2.1 Passive vs. Active Proxy

The first factor describes how actively the proxy is used in the visualization process. **Active proxy** is when the physical scale models can be or are meant to be actively manipulated by the user. **Passive proxy** is when the scale models are mostly static. This approach heavily emphasizes visualizing the spatial arrangement correctly according to the actual referent objects. The degree of association between the data representation and the active proxy can be seen and is reflected in the type of interaction modality. Consider interactive situated visualizations using referent proxies. In completely passive proxy approaches, the data manipulation is performed by separate input devices such as touch screens. For example, to select data attributes or to change the color mapping, the user interacts with GUI elements on a screen. Moving towards a completely active proxies approach, the use of external input devices is minimized. In an extreme case, the manipulation of data

representation is performed purely by physically manipulating the scale models. This arguably increases the binding between the data and physical referent. Of course, there are also middle-ground cases where the combination of physical manipulation and conventional input devices are used.

2.2 Spatial vs. Abstract Data Representation

The second factor describes the type of data representation. **Spatial representation** shows information that can be integrated with the model or space around the model. For example, a temperature map of internal building spaces. **Abstract representation** of data does not naturally integrate with the space. For example, time-series bar charts or line charts representing the energy consumption of campus buildings. Complex spatial representation of data is commonly presented as multiple geographical layers. It is highly coupled in space with the referent but having many layers can increase visual clutter. Abstract data representations are spatially decoupled from the referent. This means increasing the amount of presented information would not introduce the same occlusion issue as in spatial representation.

2.3 Quadrants

By using the factors discussed above, we come up with conceptual quadrants illustrated in Fig. 2.

2.3.1 Passive Proxy and Spatial Representation. In museums or public installations, scale models are commonly presented as static objects. Common use cases are city data visualizations. Several examples¹ include City of London, Nantes, Kuala Lumpur, and more. Many of these types of work are interactive. Spatial data visualization such as a road networks are projected directly on the static models. An input device, typically a tablet, is used to control the interactive features.

2.3.2 Passive Proxy and Abstract Representation. An example of this approach is the augmented scale model approach introduced in Satriadi et al. [17]. In their work, scale models of machines are placed on a table. Abstract data representations are presented as virtual dashboards or slides attached to the scale model. Navigation of the slides views was done by pressing virtual buttons, as opposed to physical manipulation of the models. Another example uses abstract representation such as textual information embedded to scale models [1].

2.3.3 Active Proxy and Spatial Representation. Two common ways to make a proxy active are by physical deformations and physical manipulations of the proxies. Physical deformation includes changing the shape of the models. Illuminating clay system [11, 12] updates the geographic data visualization (e.g., simulated water flow) based on the shape of the 3D models deformed by the user. The typical sandbox table with top-down projection is also an example of this approach. Physical manipulation relies on non-deformation actions, such as picking up, rotating, or arranging the scale models. Simultaneous world [7] demonstrates a data exploration technique with the manipulation of scale models. Spatial information is presented as data layers. In this work, the active proxy technique relies

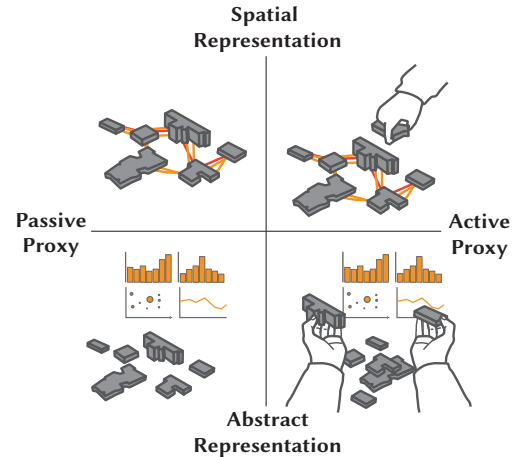


Figure 2: Conceptual quadrants showing passive versus active proxy and spatial versus abstract representation.

on the placement and arrangement of physical models in a specific interaction area. Other work that, to some extent, fall within this category include Urp [25], Uplift [4], and tangible globe AR [18].

2.3.4 Active Proxy and Abstract Representation. We observe a few cases that bind the scale models with a simple abstract data representation. For example, in the CityMatrix [28] system, the user can rearrange the city layout. The layout then updates a radar chart showing multiple city-related attributes on a large display. It is common for this type of work to conserve spatial information by using a map, which can encourage a specific arrangement of the models. While this is a good way to keep spatial information, it might discourage physical manipulations. An alternative approach that, we believe, encourages physical manipulations is by removing the map from the interaction space. The spatial information can, for example, be presented separately as a digital map.

3 ACTIVE PROXY DASHBOARD

We focus on the active proxy and abstract representation quadrant for two main reasons. First, abstract representations allow for the visualization of a large number of attributes. Second, we want to explore how active manipulations of scale models facilitate the understanding of complex data, including its association with the physical referents. To test and provide a better understanding of the feasibility of the active proxy concept, we developed a prototype system called *Active Proxy Dashboard*. Our target scenarios are prox-situated visualization [16] using multiple scale models combined with abstract representations of multivariate data.

3.1 Physical Referent Proxies

Proxies in our prototype are physical scale models of campus buildings. We printed seven hand-held-size models. These proxies are placed on a standing coffee table placed in front of a display. In reality, there are many buildings on the campus. As proof of concept, we only use models of essential buildings. One of the key concepts is to encourage the physical manipulation of scale models. Thus,

¹<https://nla.london/videos/new-london-model>, http://devocite.com/?page_id=1202, <https://cubexis.com/my/portfolio/scale-model-projection-mapping/>

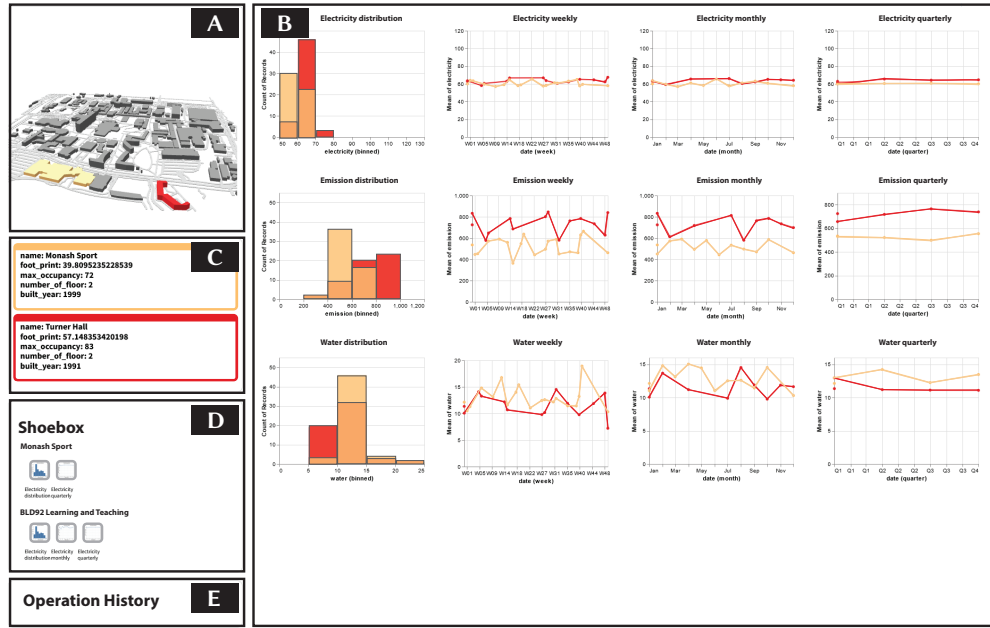


Figure 3: Dashboard of our implementation of Active Proxies consists of the 3D map view (A), charts panel (B), building information (C), shoebox panel (D), and operation history panel (E). In this screenshot, two proxies are picked up, triggering charts filtering.

we let the models be scattered on the table. The spatial information is preserved on the dashboard described in the next section.

3.2 Dashboard

The data representations are displayed on a Microsoft Surface Hub. Details are as follows.

3.2.1 Spatial Information. Our approach to active proxy is to move away from the rigid arrangement of the scale model according to the campus map. We expect that this will encourage the user to grab and manipulate the scale models. This also means that the scale models do not reflect the spatial information of the actual referents. To keep this information, we provide a 3D map view to provide the overview of the campus as well as to indicate the location of the building represented by the scale model that is being picked up (Fig. 3-A).

3.2.2 Data Representation. The charts are an essential component of the active proxies approach. In our implementation, we show 12 charts derived from three temporal attributes: electricity consumption, emission, and water consumption (Fig. 3-B). Each row represents a single attribute presented at different levels of detail (e.g., histogram of distribution, and line charts of monthly trend, weekly trend, and quarterly trend).

3.2.3 Descriptive Information. We provide descriptive information on each building. This could be quantitative information such as the area of the building, maximum occupancy, number of floors, or built year. Qualitative information such as brief history or other textual descriptions can also be shown in the dedicated information panel (Fig. 3-C).

3.2.4 Shoebox. As part of the sensemaking process, there is a concept known as “shoebox”, which involves filtering and storing useful information from a vast information domain [13]. While the dashboard presents a large amount of information and an overview, during the data analysis process, it is common to focus only on specific attributes.

We support the shoebox approach by allowing the user to select certain charts for each referent. Each referent can have a different set of charts. For example, energy consumption is relevant for the Teaching and Learning building with extensive air conditioning systems while water consumption is more relevant to the Sports Building with swimming pools. The information stored in the shoebox panel is visualized in the dashboard (Fig. 3, D). We also made this information persistent so that it can be reused throughout multiple analysis sessions.

3.3 Physical Manipulations

We demonstrate several physical manipulation-related actions. We extend the type of physical manipulation in existing work by exploring in-air manipulations. In the current prototype, we used the following actions. 1) **Picked up** is when a proxy model is picked up from the table. 2) **Air dwell** is when a picked-up model is held stationary in the air. 3) **Placed on the table** is when a picked-up model is placed back to the table. 4) **Enter a zone** is when a scale model enters a pre-defined three-dimensional space. 5) **Dwell in a zone** is when a scale model dwells in a pre-define three-dimensional space. 6) **Exit zone** is when a scale model exits a pre-defined three-dimensional space. 7) **Roll and pitch** are when a scale model is

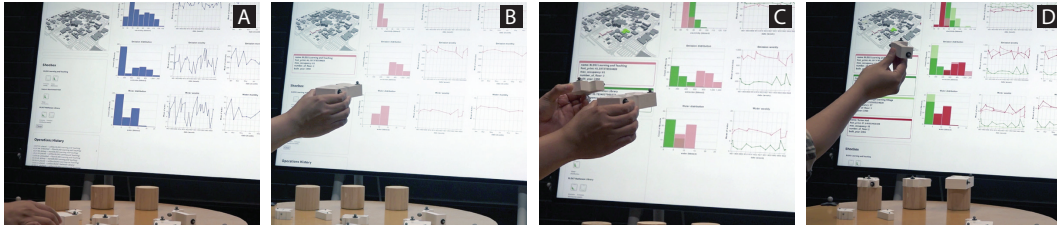


Figure 4: Filtering process in our active proxy dashboard. A) overview is the default state. B) The user picked up a model, triggering global filtering. This filtering is also applied when the user picks up two models with two hands (C) or more than two models with the help of mini pedestals (D).

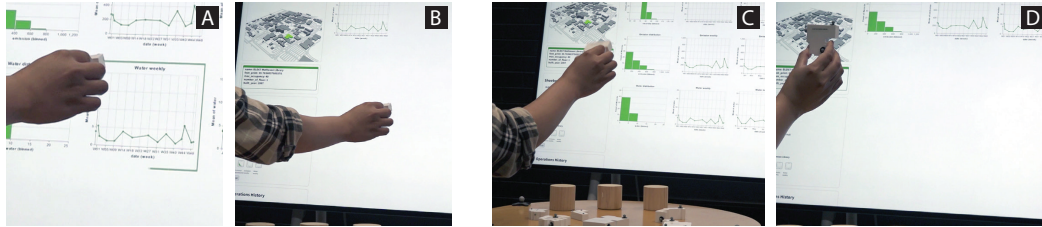


Figure 5: Chart selection process in our active proxy dashboard begins by moving and dwelling a model in the proximity of the intended chart (A). After a short dwelling time, that chart is stored in the shoebox and focused (B). Moving the model away from the chart brings back all default charts (C). Pitching the model shows charts in the shoebox that are associated with the model (D).

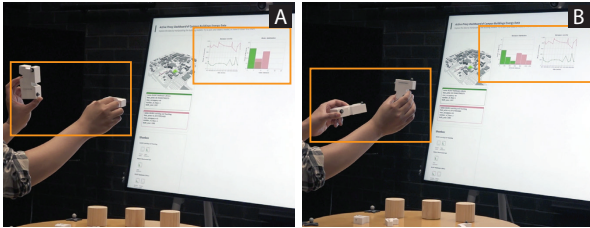


Figure 6: Filtering can be combined with the charts focusing. For example, picking up two models and pitching one will filter and focus the charts. Visible charts depend on which model is pitched. A) Selected charts of the model in the left hand are focused. B) Selected charts of the model in the right hand are focused.

rotated. We did not consider yaw in our initial prototype since this rotation can be arbitrary when the user picks up the model.

3.4 Technical Details

We implemented a web-based dashboard prototype using vegalite [19] and map view using the three.js library. We use a Vicon system to track models instrumented with passive markers. The tracking data is sent through an interpreter built using Unity that detects the physical manipulation events. Events are sent to the dashboard via WebSocket. The web server stores the mapping between physical manipulation events (e.g., picked up) and data representation operations (e.g., filter) in a JSON file. The dataset is stored

as a CSV file in the dashboard’s server. We placed the scale models on a round standing table in front of the dashboard, as shown in Fig. 1-middle.

3.5 Initial Workflow Design

There are many ways to pair physical manipulations and visualization operators. At this early stage, we designed a data exploration workflow that supports three key goals: 1) overview, 2) filter and comparison, and 3) select and focus.

The *overview* goal is to provide high-level information of the data across all buildings. The default state of the dashboard—when all models are placed on the table—shows the overview data (Fig. 4-A). In this state, all charts show aggregated values. For instance, the monthly electricity usage trend shows mean values across all buildings in the data.

The *filter and comparison* goal is the ability to inspect data of a specific building by filtering the charts. This is mainly achieved by pick-up and air dwell physical manipulations. Picking up a scale model can be a clear indication of interest. Therefore, it makes sense to map this physical manipulation to the focus-related operators such as filtering the entire charts on the dashboard. When a model is picked up, the charts are filtered by this referent (Fig. 4-B). We highlight the referent on the map view as well as show the general information of the referents on the building information panel. This filtering interaction supports multiple referents (Fig. 4-C), allowing for data comparison between referents. We also provide small pedestals placed on the table, akin to the virtual pedestal metaphor in previous work [29], on which the scale models can be

rested to support more than two filters (Fig. 4-D). Placing back the scale model removes it from the filter list, which eventually brings back the overview when the list is empty.

The *select and focus* goal is related to the shoebox feature mentioned in section 3.2, where we allow the user to select specific charts (to be placed in the shoebox) for a specific referent proxy. The selection is performed by picking up a model and then moving it to near the intended chart (Fig. 5-A). After a short dwell time, the chart is then selected, copied and stored in the shoebox, and focused (Fig. 5-B). Moving the model away from the chart brings back all default charts (Fig. 5-C). To focus on previously selected charts, the user needs to pick up and pitch the scale models (Fig. 5-D). This select and focus interaction can also be combined with filter and comparison, as illustrated in Fig. 6-A and B. See the supplementary video for further demonstration of all interactions.

4 REFLECTION AND FUTURE WORK

Situated visualization work has been heavily focused on creating a binding between the physical referent proxy and data representation through spatial grouping (i.e., placing charts close to the proxy). In proxysituated scenarios, in particular, those that are using scale models as referent proxies, the tactility of these models is rarely utilized in the design as the binding is mainly performed by space. We entertain the possibility of exploiting the tangible interaction of the proxy to build binding between data representation and referent in the user's mind. Our initial exploration, including prototyping, of such active proxy concepts has led to several key points for future research.

The most important future work is to test the tangible interaction binding hypothesis. We plan to investigate how well the users' learning performance with the active proxy dashboard is compared with that of passive approaches. Examples of possible passive techniques include direct manipulation of charts using touch gestures. The study will then provide insights into the first question we asked in the introduction. Can tangible interactions create a strong binding between physical referent and abstract data representation in the user's mind?

Other future directions revolve around the second question, which is improving the active proxy design. We identify five potential future directions to address this question. 1) One possible approach is introducing storage space and interactions space. Storage space is where all models are initially placed. The user picks up models that they are interested in and moves them to the interaction space for exploration. A tabletop map could be used as the storage space to arrange the models according to the real-world layout. 2) Another consideration includes dealing with large relative differences. Some buildings are much larger than others. As discussed in the previous work [16], the scale of the proxy needs to be considered and in active proxy scenarios, it could mean sacrificing the relative differences over the ability to easily manipulate the models. 3) It would be interesting to see the effect of mixing spatial binding and tangible binding, for instance, by embedding virtual charts to active proxy models in augmented reality. 4) There are potential design aspects in collaborative settings, such as the users' territoriality [21], workspace awareness [6], and communication cues [14]. 5) In addition to collaborative scenarios, such as for

community engagement and education, we also envision in-depth analysis use cases of activity proxies in personal workspaces (Fig. 1-right).

To better understand the benefits of situated visualization using referent proxies it is important to develop a theoretical framework to describe bindings between physical referents and data presentations. From the application end, it is also important to test such systems in the wild. We plan to work with domain experts to improve our active proxy dashboard prototype. Currently, the prototype relies on a motion capture system. While this setup ensures tracking precision and stability, its complexity and cost could be a barrier to its democratization. Future technical improvements should investigate more lightweight solutions such as utilizing the webcam for computer vision-based object tracking.

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